
RESEARCH INTERESTS

I am broadly interested in Computational Linguistics and Natural Language Processing, specifically (1) Cultural alignment of LLMs, (2) Developing robust evaluation frameworks, and (3) Investigating model internals to advance LLM interpretability and AI safety.

EDUCATION

Jahangirnagar University

Dhaka, Bangladesh

*Master of Science in Computer Science**Jul. 2026 (Expected)*

- **Current CGPA:** 3.97/4.0
- **Relevant Coursework:** Machine Learning (A+), Deep Learning (A+), Natural Language Processing (A+), Data Mining (A), Artificial Intelligence & Expert Systems (A+), Data Structures & Algorithms (A+), Software Engineering & Database Systems (A+), Computer Architecture & Operating Systems (A+), Programming Languages (A+)

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh

*Bachelor of Science in Civil Engineering, with Honors**Mar. 2025*

- **CGPA:** 3.77/4.0, **Major:** Transportation Engineering, **Minor:** Environmental Engineering
- **Relevant Coursework:** Differential Equations & Statistics (A+), , Computer Programming Sessional (A+), Engineering Computation Sessional (A+), Differential, Integral Calculus and Matrices (S) [†]

[†]Satisfactory; graded S/F in 2020-21 due to COVID-19.

RESEARCH EXPERIENCE

Graduate Researcher

Aug. 2025 – Present

*JU CIDL Research Lab**Supervisor: Dr. Md Musfique Anwar*

- **Event Detection in Bangla Text: Assessing Language Model Robustness to Noisy Data**

Research Project, Manuscript in preparation

- * Developed an annotated Bangla event detection dataset and evaluation framework to assess the performance of instruction-tuned LLMs and encoder-based models on noisy, ASR-transcribed news text.
- * Created perturbed datasets to investigate model robustness and characterize performance degradation under varying levels of Word Corruption Rate.

- **BanglaSocialBench: A Benchmark for Evaluating Sociopragmatic and Cultural Alignment of LLMs in Bangladeshi Social Interaction**

Jan 2026 – May 2026

Tanvir Ahmed Sijan, S. M Golam Rifat, Pankaj Chowdhury Partha, Md. Tanjeed Islam, Md. Musfique Anwar*Accepted (Oral) at ACL Student Research Workshop (SRW) 2026 [\[arXiv\]](#)*

- * Conceptualized and led the development of BanglaSocialBench, an annotated benchmark comprising 1,719 culturally grounded instances across three domains for evaluating the sociopragmatic competence of large language models.
- * Benchmarked 12 LLMs in zero-shot setting, uncovering over-politeness, religious kinship term conflation, and rigidity in ambiguous social contexts.

Undergraduate Thesis

Nov. 2023 – Feb. 2025

*Bangladesh University of Engineering and Technology (BUET)**Supervisor: Dr. Md. Shamsul Hoque*

- **Assessing the Visual and Built Environment Risk on Pedestrian-Vehicle Crash Severity in Absence of Detailed Roadway Inventory: A Deep Learning Based Framework Harnessing Public Geospatial Data**

Tanvir Ahmed Sijan, Md Asif Raihan, Sk. Md. Mashrur, Md. Shamsul Hoque*Submitted to Accident Analysis & Prevention, 2025*

- * Built an end-to-end multimodal data pipeline integrating Google Street View API, Overture Maps, OpenStreetMap, and crash datasets to generate urban risk features for pedestrian crash severity prediction.

- * Applied semantic segmentation (Detectron2 + Mask2Former) on 12K+ street-view images and trained ensemble ML models (XGBoost, Random Forest, AdaBoost, CatBoost) with SHAP-based interpretability to identify key environmental risk factors.

WORK EXPERIENCE

Maks Inc.

Dhaka, Bangladesh (HQ: Frisco, TX, USA)

Research Intern

May 2025 – Sep 2025

Projects: Study on Heavy-Duty Vehicles (**World Bank Group**), Akhaura–Tamabil Land Port Feasibility Study (**Asian Development Bank**)

- Analyzed large-scale BRTA fleet data to model vehicular emissions, performed spatial analysis with QGIS, conducted nationwide stakeholder interviews, and synthesized findings for policy-oriented technical reports.

HONORS & AWARDS

Undergraduate Student Research Grant

Jan. 2024

Research and Innovation Centre for Science and Engineering, BUET

Dean's List Scholarship

Jul. 2019

Bangladesh University of Engineering and Technology (BUET)

- Awarded for maintaining an average GPA of 3.75 or above across two consecutive regular terms.

3rd Place, Concrete Solutions Competition

Mar. 2022

American Concrete Institute (ACI)

- Recognized for the most innovative team design; project featured in the **Daily Prothom Alo** [\[Feature\]](#)

Higher Secondary Certificate (HSC) Scholarship

2019

Board of Intermediate and Secondary Education, Rajshahi

- Awarded **Talentpool Scholarship**; ranked **10th (top 0.01%)** among all candidates in the Rajshahi Board.

LEADERSHIP & VOLUNTARY WORKS

ITE BUET Student Chapter

Dec. 2024 – Mar. 2025

Deputy Vice President

- Collaborated with other executive members to organize workshops and seminars on various topics of transportation research.

ACI BUET Student Chapter

Mar. 2024 – Mar. 2025

Vice President

- Organized poster competitions, seminars, and field visits in coordination with chapter members.

SKILLS

- **Programming:** Python, C, C++, SQL, MATLAB, LaTeX
- **Libraries:** NumPy, PyTorch, Transformers, Matplotlib, Scikit-Learn, Selenium
- **Other:** Git, LaTeX, Bash
- **Languages:** Bangla (Native), English, Hindi (Basic)